

Analysis of Coupling Effect in Coupled SOI Rib Waveguide for Different Polarization Mode

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Abstract — Parallel coupled Rib waveguide based on silicon-on-insulator (SOI) has been explored to analyze the coupling effect on different polarization mode using the simulation approach based on the finite element method and the semi-vector finite difference beam propagation method. The minimum coupling effect has been analyzed with respect to variation of geometry parameter and comparison of different polarization modes (E_{11}^x and E_{11}^y) in parallel Rib waveguide. The coupling effect has been observed with respect to the variation in waveguide spacing (S) for some fixed value of ratio of slab height to total Rib height (i.e., $r = h/H$), such as 0.6, 0.7, and 0.8 under single mode propagation constraint at operating wavelength of 1.55 μm . However, the coupled Rib waveguide with $r = 0.6$ at the different waveguide spacing is most suitable for the minimum coupling effect and for the enhancing the data transmission capacity.

Keywords—Rib waveguide, SOI material, Coupling length, Coupling coefficient.

I. INTRODUCTION

In the integrated photonic system, coupled optical waveguide is one of the most essential components to enhance the data transmission capacity for single mode operation in two parallel Rib waveguides. The remarkable optical properties of silicon-on-insulator (SOI) material over other materials used for integrated optical waveguide are enhanced confinement factor and low propagation loss due to high index contrast between the core and cladding/substrate at the operating wavelength [1].

The optical power coupled from one waveguide to the neighbouring waveguide by the phenomena of mode coupling, which can be described using the coupled mode theory (CMT) or coupled power theory (CPT) [1, 11]. Further, to explore the coupling behavior by using the coupled mode theory, several factors related to the optical waveguide, such as waveguide spacing, the geometry of waveguide and the operating wavelength must be taken into considerations [3]. The main aim for the current research work is to reduce the coupling effect, minimizing the losses and enhance the data capacity in suitable single-mode large cross-section Rib waveguide structures. The transmission capacity can be enhanced by the use of multimode waveguide, but it is having some problems like, mode interference, losses at higher order modes, etc. The use of two single mode parallel Rib waveguide can overcome the problems associated with multimode optical waveguide to a great extent. The geometry of optical waveguide has significant impact on the design of coupled Rib waveguide structure for single-mode operation. Therefore, current

research also includes the analysis of coupling behavior for three different geometries depend on ratio of slab height ' h ' to total Rib height ' H ' (i.e., $r = h/H$).

Recently, the several method have been used for analysis of coupling behavior such as, Finite element method (FEM), Finite difference beam propagation method (FDBPM) and its modifications, etc. [1-3, 13, 14]. In this paper, simulation based FEM and FDBPM have been used for the analysis of coupled homogeneous optical Rib waveguides, in order to investigate the mode coupling between parallel Rib waveguide for different values of $r (= h/H)$.

II. DESIGN ANALYSIS

Among the various possible optical waveguide structures, such as, strip waveguide, strip loaded waveguides, buried waveguide, Rib waveguide, etc., the Rib waveguide has been chosen due to its significantly low propagation losses [3, 10] and the analysis based on coupling effects, etc. has been done. The refractive index of substrate (n_2), which made up of SiO_2 and that of the core (n_1), which is made up of silicon are 1.5, and 3.5, respectively. Air is used as an upper cladding region, having refractive index (n_0) as 1. The cross sectional view of the Rib waveguide is shown in Fig. 1, where, total Rib height (H) and Rib width (w) are fixed as 5 μm , while the Rib depth (d) and slab height (h) is varied depending on the value of ratio ' r ', for single-mode propagation condition.

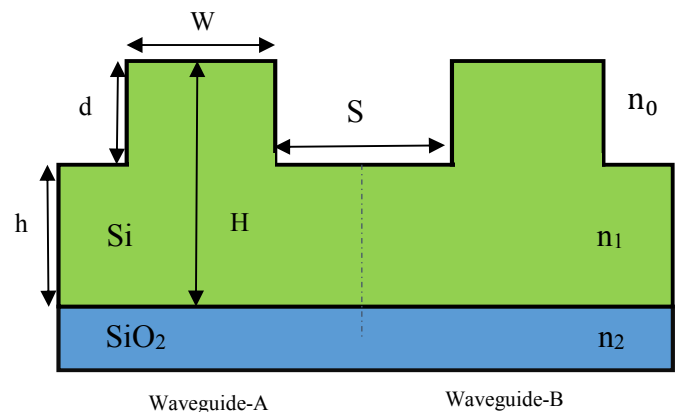


Fig. 1. Cross-section of two parallel symmetrical Rib waveguide.

In few cases, the mode-division multiplexing (MDM) based systems, where the multimode waveguides are utilized to increase channel numbers, have been reported [8]. Usually this type of waveguide is having the problem of mode interference within the core structure that can be overcome by the use of two parallel single mode Rib waveguide.

From last few decades, many authors have focused on the study of single mode condition in Rib waveguide [4-7]. Firstly, the single mode condition in Rib waveguide was proposed by Soref [4] and which was modified by Powell [6], Pogossian [7], etc. The condition for single mode in Rib waveguide as described by Powell is re-written in Eq (1) as,

$$\frac{w}{H} \leq 0.3 + \frac{r}{(1-r^2)^{\frac{1}{2}}} \quad (1)$$

where, $r = \frac{h}{H} \geq 0.5$

For coupling analysis in Rib waveguide, different geometries have been considered to satisfy the single mode condition, and is noted in Table 1.

TABLE I. DESIGN GEOMETRY OF RIB WAVEGUIDE AT SINGLE MODE CONDITION

Structure No.	Rib width (W) (μm)	Total Rib height (H) (μm)	Slab height (h) (μm)	Rib depth (d) (μm)	Ratio ($r = h/H$)
1	5	5	3.0	2.0	0.6
2	5	5	3.5	1.5	0.7
3	5	5	4.0	1.0	0.8

III. MODE ANALYSIS AND SIMULATION

The modal field distribution of the y-component of the electric field for the dominant odd and even modes have been calculated through the finite element method based Rsoft FemSIM software and shown in Fig. 2. The effective refractive indices for fundamental even, and odd modes are 3.494363, and 3.494347 respectively. Here, the spacing between waveguides is assumed to be $S = 6 \mu\text{m}$ and ratio (r) is of 0.6 at operating wavelength ($\lambda = 1550 \text{ nm}$).

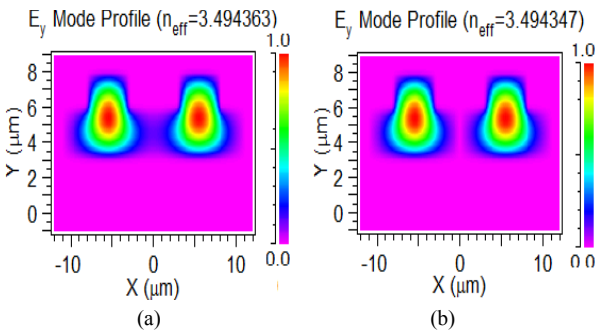


Fig. 2. Mode profile of the coupled Rib waveguide at $S = 6 \mu\text{m}$ for (a) even mode (b) odd mode

The field amplitude distribution pattern of above mode profiles are shown in Fig. 3, where the waveguide structure width is on the x -axis and the mode amplitude is on the y -axis. In the even mode, the mode amplitude distributions are in the same phase, whereas in the odd mode, the mode amplitude distributions are in opposite phase. Further, the

simulation (using FDBPM based OptiBPM software) results of designed coupled Rib waveguide in terms of the optical field intensity distribution along waveguide length and the corresponding normalized power distribution is shown in Fig. 4(a), and 4(b) respectively. Here, the input power is launched in one of the waveguide B, and then the total power is coupled in the adjacent waveguide A after the certain length (known as coupling length ' L_c ', where power is fully coupled in waveguide A). The direction of field propagation is considered in z -direction and the width of the Rib waveguide is considered in x -direction.

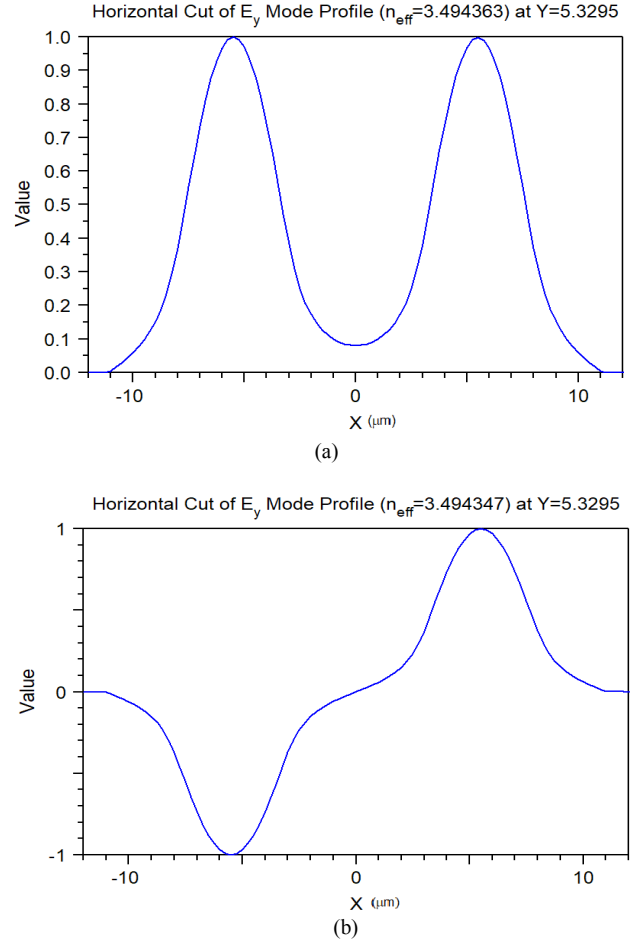
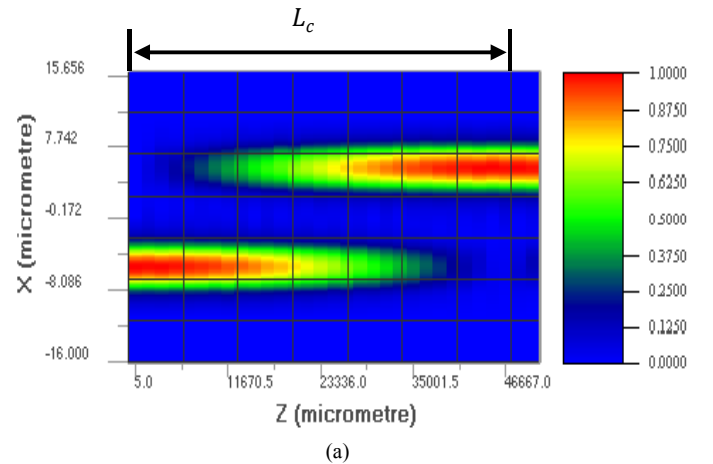


Fig. 3. Field amplitude distribution of mode (a) even mode (b) odd mode.



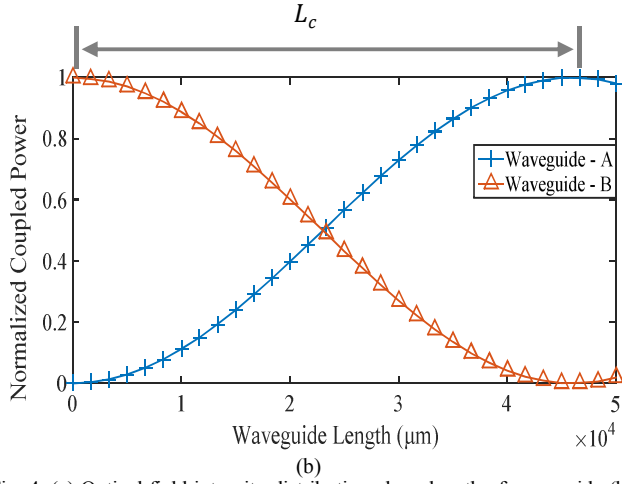


Fig. 4. (a) Optical field intensity distribution along length of waveguide (b) variation of optical power along length of waveguide.

IV. RESULTS ANALYSIS

The full vector finite element method is one of the suitable method for the calculation of mode propagation constants (even/odd modes) and the modal profile of the coupled Rib waveguide. The mode propagation constant (β) for even and odd modes of the structure have been calculated using the finite element method based FemSIM software, which further utilized to estimate the coupling length of the coupled Rib waveguide structure. The coupling of mode and coupling length ' L_c ' of coupled waveguide structure has been shown in Fig. 4. The coupling length of parallel straight Rib waveguides can be defined in term of even and odd mode propagation constants [10], as

$$L_c = \frac{\pi}{(\beta_e - \beta_o)} \quad (2)$$

The coupling length of the guided modes of two parallel Rib waveguide have been calculated by varying the waveguide spacing for the fundamental polarization mode with different value of ratio ' r ', and shown in Fig. 5. Moreover, the coupling length and coupling coefficient of the guided modes, as shown in Figs. 6, and 7 respectively, have been calculated and compared for different polarization modes (E_{11}^x and E_{11}^y) at different value of ratio ' r ' using the Rsoft FemSIM tools. The figures show that the increase in the waveguide spacing ' S ' leads to the increase in coupling length and decrease in mode coupling effect.

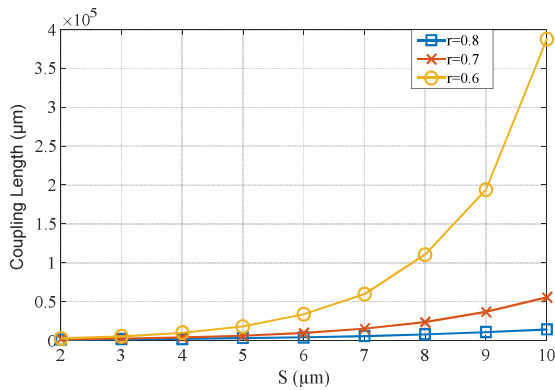


Fig. 5. Coupling length vs. waveguide spacing (S) at different value of ratio ' r ' for x-component of the fundamental mode.

Fig. 6 shows the relationship between coupling length (L_c) and the waveguide spacing ' S ' for different value of ratio ' r ' for different polarization modes. The results show that the minimum coupling effect can be achieved with the E_{11}^y mode for each considered value of r .

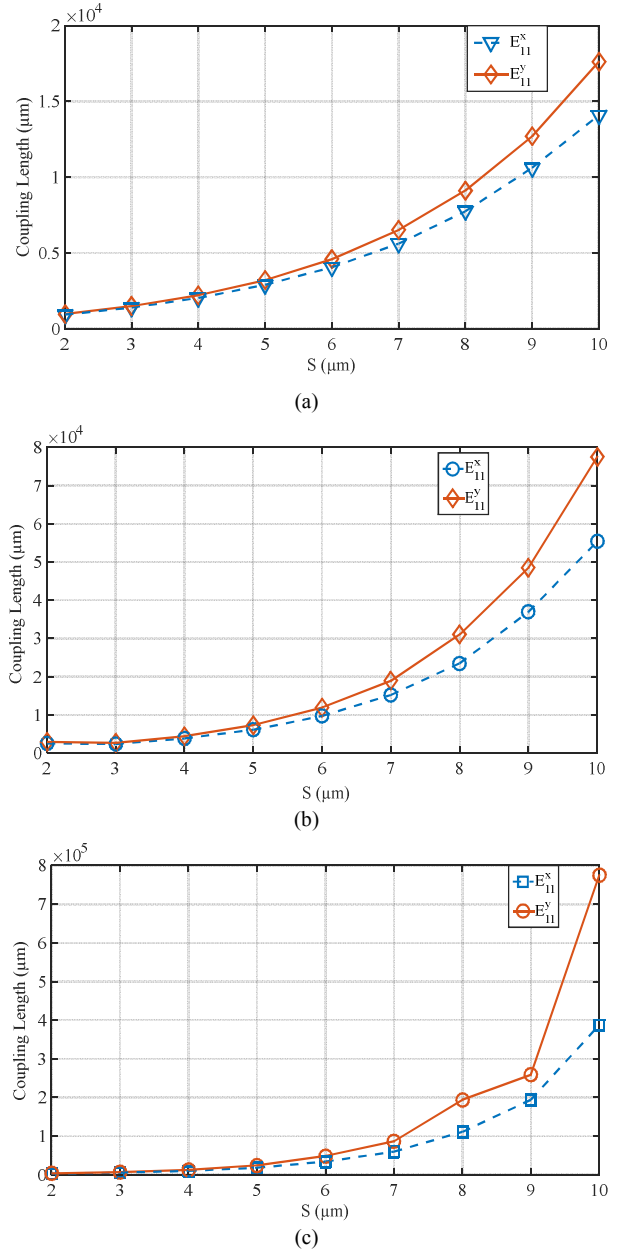


Fig. 6. Coupling length vs. waveguide spacing (S) with fixed value of ratio ($r = h/H$) in E_{11}^x and E_{11}^y polarization (a) $r = 0.8$ (b) $r = 0.7$ and (c) $r = 0.6$.

Fig. 7 shows the dependency of coupling coefficient on the ratio ' r ' for different waveguide spacing (S) for the coupled Rib waveguide structure. The solid line is for E_{11}^x mode, and the dash line is E_{11}^y mode. The E_{11}^y modes have lesser mode coupling effect compared to E_{11}^x mode in each of the considered case of ratio ' r '. The mode-coupling coefficient (K) can be obtained for homogeneous structure, using the following expression as provided in [2, 13]

$$K = \frac{\pi}{2L_c} = \frac{\beta_e - \beta_o}{2} \quad (3)$$

It clear from the result that the minimum coupling coefficient and larger coupling length for the design presented can be achieved at ratio value (r) of 0.6 under the constraint of single mode condition. During the simulative analysis, it was also observed that for ratio ' r ' < 0.6, the propagation may not remain as single mode but it may be multimode propagation with increased power loss. Therefore, the minimum value of ratio ' r ' = 0.6 is more selective value in single mode Rib waveguide structure from the point of view of minimum propagation delay [12], mode coupling effect, etc.

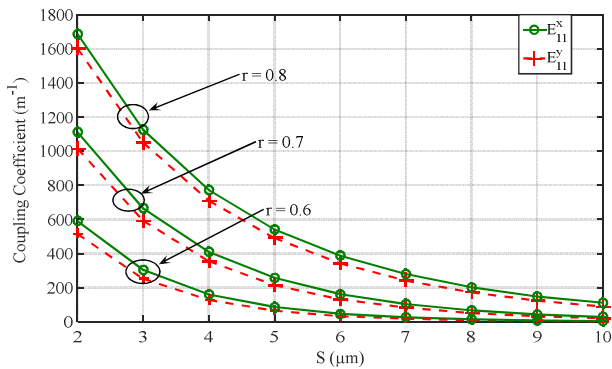


Fig. 7. Coupling coefficient vs. waveguide spacing(s) at different value of ratio(r) for different polarization mode.

The coupling analysis has also been performed with the finite difference beam propagation methods, such as, semi-vectorial (TE), semi-vectorial (TM), etc. using the OptiBPM software. The simulation result for coupling length with respect to the waveguide spacing variations by different methods at the fixed value of r ($= 0.6$) has been done, and shown in Fig. 8. The figure shows that the semi-vectorial (TM) method provides the minimum coupling effect for the design presented. Although, the scalar approximation method is simple and accurate for the modal field and coupling length estimations, but here the semi-vectorial method, especially, semi-vectorial (TM) is more accurate method for analysis of coupling properties.

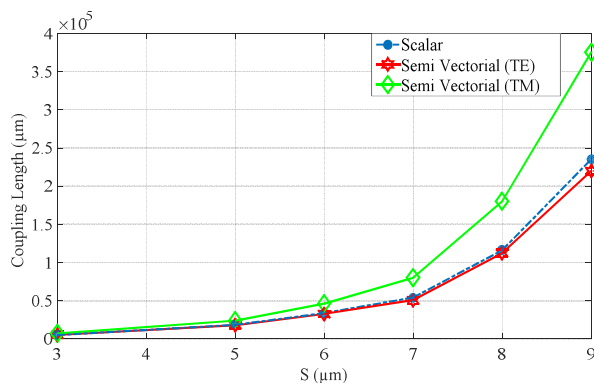


Fig. 8. Coupling length of waveguide spacing (s) in different method with fixed value of ($r = 0.7$).

V. CONCLUSION

Design of parallel coupled optical Rib waveguides to reduce the coupling effect and to enhance the data

transmission capacity under single mode condition has been proposed. The coupling analysis of coupled structure based on SOI guiding layer has been done by the finite difference beam propagation method (FDBPM) and finite element method (FEM). The coupling effect has been tested for three different values of ratio ' r ' of slab height (h) to total Rib height (H) (i.e., $r = 0.6, 0.7$, and 0.8). The lowest coupling effect can be achieved with the minimum value of ratio of slab height (h) to total Rib height (H) (i.e., $r = 0.6$). It was also observed that at different value of r , the coupling length in different polarization mode increases with the increase in waveguide spacing. The coupling effect has also been simulated with the finite difference beam propagation methods, such as, semi-vectorial (TE), semi-vectorial (TM), etc. The comparative analysis shows that the minimum coupling effect for the presented design can be achieved with the semi-vectorial (TM) method.

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